Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 2 – Maclaurin Series

Medium & Hard • 25 Questions

Scan for free resources

Section A – Medium (Q1–15)

1. State the Maclaurin series for e^x. **[2]**

2. State the Maclaurin series for $\sin x$. **[2]**

3. State the Maclaurin series for $\cos x$. **[2]**

4. State the Maclaurin series for $\ln(1+x)$. **[2]**

5. Find the Maclaurin series for e^{2x} up to x^3. **[3]**

6. Find the Maclaurin series for $\sin 2x$ up to x^3. **[3]**

7. Find the first three non-zero terms of the Maclaurin series for $\cos 3x$. **[3]**

8. Find the Maclaurin series for $\frac{1}{1-x}$ up to x^3. **[3]**

9. Differentiate the series for $\sin x$ to obtain the series for $\cos x$. **[3]**

10. Integrate the series for $\frac{1}{1+x}$ to obtain the series for $\ln(1+x)$. **[3]**

11. Find the Maclaurin series for $(1+x)^{-1/2}$ up to x^2. **[4]**

12. Use the series for e^x to approximate $e^{0.1}$ to 4 d.p. **[3]**

13. Find the range of validity of the series for $\ln(1+x)$. **[2]**

14. Find the Maclaurin series for $x\cos x$ up to x^3. **[3]**

15. State the general form of a Maclaurin series. **[2]**

Section B – Hard (Q16–25)

16. Find the Maclaurin series for $\tan x$ up to the term in x^3. **[5]**

17. Find the Maclaurin series for $e^x \sin x$ up to x^3. **[5]**

18. Find the first four non-zero terms of the series for $\ln(\cos x)$. **[6]**

19. Use Maclaurin series to evaluate $\lim_{x\to 0} \frac{\sin x - x}{x^3}$. **[4]**

20. Find the Maclaurin series for $\operatorname{arsinh} x$ up to x^3. **[5]**

21. Find the series expansion of $\frac{1}{\sqrt{1+x^2}}$ up to x^4. **[5]**

22. By integrating a suitable series, find a series for $\arctan x$. **[4]**

23. Find the Maclaurin series for $e^{\sin x}$ up to x^4. **[6]**

24. Determine the radius of convergence of the series for $\ln(1+x)$. **[3]**

25. Use series to approximate $\int_0^{0.2} \frac{\sin x}{x}\,\mathrm{d}x$ to 4 d.p. **[5]**

End of Worksheet • *Show all working clearly.*